LEAP hub and Grafana are down

Status: Reviewed

Overview

The hub and Grafana were not reachable because of a core node that ran out of TCP memory. Even though the hub and grafana pods were running without issues, the pods on the problematic nodes were failing, including kube-dns, which made the hub and Grafana unreachable networking wise.

The system eventually self-healed and the hub and Grafana showed up again.

What Happened

A core node ran out of TCP memory, which made the kube-dns pod unable to resolve DNS queries.

The pod tried to restart a few times but failed. During this time frame (~30m) the hub and Grafana were not reachable.

According to logs and pods age, the hub and the prometheus never stopped running during this time frame.

Resolution

It wasn't until the node was killed and kube-dns was scheduled on another node that the issue was fixed.

Where we got lucky

An user reported the problem and the system self-healed in about 30m.

What Went Well?

The system self-healed in about 30m and we were able to take the time to find the issue.

What Didn't Go So Well?

The community reported the issue to us.

Logs exporter showed the TCP OOM error as an INFO message, which made it really hard to spot.

Action Items

- https://github.com/2i2c-org/infrastructure/issues/3299
- kubelet should track tcp_mem stats also along with cpu/ram/disk as tracked in upstream kubernetes repo https://github.com/kubernetes/kubernetes/issues/62334

Timeline

OWNER OF REVIEW PROCESS

Georgiana

IMPACT TIME

Sep 20 at 17:18 to Sep 21 at 15:57

DURATION

22h 39m 49s

*All times listed in this report are in Bucharest.

Sep 20, 2023

5:18 PM

Triggered by Georgiana through the website.
Description: LEAP hub and Grafana are down (View Message)
INCIDENT #262
LEAP hub and Grafana are down

5:19 PM Cluster events showed messages about `node/gke-leap-cluster-core-pool-1592e1d3-gnks` being healthy

Running `k get events` shows the following events related to `node/gke-leap-cluster-core-pool-1592e1d3-gnks`:

- 113s Normal NodeNotReady
- 2m48s Normal NodeHasSufficientMemory
- 2m48s Normal NodeHasNoDiskPressure
- 2m48s Normal NodeHasSufficientPID
- 2m48s Normal NodeReady

5:31 PM By this time, the hub and Grafana became available again and the outage ended

Georgiana Dolocan #managed_jupyt_inc_262
Ok, for some reason the hub and grafana are up again :confused: I'm poking around the console to figure out what happened

5:47 PM

Georgiana Dolocan #managed_jupyt_inc_262
I'm seeing bunch of error in the `support-ingress-nginx-controller` pod,
Also, the `support-ingress-nginx-controller` and the `support-grafana` pods seem to have been restarted around the time the hub and grafana came up again.

6:47 PM

Georgiana Dolocan #managed_jupyt_inc_262
Also, in grafana there's a 2:30h gap in data

7:08 PM

Georgiana Dolocan #managed_jupyt_inc_262
My only explanation so far is that the something happened with node `gke-leap-cluster-core-pool-1592e1d3-gnks` which made all of the pods on it restart before the node died and the pods were rescheduled on a new one:

7:48 PM

Georgiana Dolocan #managed_jupyt_inc_262
I've seen a weird message about the reason why this node poll was deleted `"Deleting node gke-leap-cluster-core-pool-1592e1d3-gnks because it does not exist in the cloud provider".`

7:55 PM

Urgency changed from high to low by Georgiana.
INCIDENT #262
LEAP hub and Grafana are down

8:08 PM

Georgiana Dolocan #managed_jupyt_inc_262
Ok :slightly_smiling_face: I think I will logout and try to wind off.

My conclusion atm is that the hub was never down, just unreacheable because of the kube-dns pod restarting on another node. Also, the time it took to figure out that the old

one was not healthy was pretty long and it kept appearing as ready in the logs, though nor the hub or grafana were reacheable during that interval :confused:

8:09 PM

yuvipanda #managed_jupyt_inc_262

yeah that makes sense. 'node failure'

Sep 21, 2023

3:57 PM

Resolved by Georgiana through Slack.

INCIDENT #262

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3:57 PM

Note added by Georgiana.

Resolution Note: I've concluded that the Hub and Grafana weren't in fact down, but just not reacheable because of a node failure that caused the kube-dns pod to not be able to recover after a crash until the faulty node was deleted and the pods restarted on a new, healthy one.

The reason for the crash is still unknown or why the exiting node failed to accommodate the restarted kube-dns pod. Relevant logs can be found in the dedicated slack channel of this incident.

INCIDENT #262

LEAP hub and Grafana are down

8:26 PM

yuvipanda #managed_jupyt_inc_262

@georgiana.dolocan i wonder if the node just ran out of RAM?

Sep 22, 2023

1:44 PM

Georgiana Dolocan #managed_jupyt_inc_262

YES! I think I finally managed to found the error (which is an info message :face_with_rolling_eyes::disappointed:

```{
"textPayload": "[16135260.261524] TCP: out of memory -- consider tuning tcp_mem\\r\
\n",
"insertId": "0",
"resource": {
"type": "gce_instance",
"labels": {
"project_id": "leap-pangeo",
"zone": "us-central1-c",
"instance_id": "5110376345040945098"
}
},
"timestamp": "2023-09-20T13:54:18.318927329Z",
"severity": "INFO",
"labels": {
"http://compute.googleapis.com/resource_name": "gke-leap-cluster-core-pool-1592e1d3-gnks",
"is_text_escaped": "true"
},
"logName": "projects/leap-pangeo/logs/serialconsole.googleapis.com%2Fserial_port_1_output",
"receiveTimestamp": "2023-09-20T13:54:20.328247998Z"
}```
```

This message was logged for ~30m until showing other kernel errors and then finaly killing the node.

2:06 PM

Georgiana Dolocan #managed_jupyt_inc_262

Aha, there's an open issue about kubelet not monitoring the tcp mem of the node, but only disk ram and cpu and this is why node appeared as ready for more than 30m even though the tcp oom was happening https://github.com/kubernetes/kubernetes/issues/62334.

And this is how network traffic on the faulty node looked like just before the outage:

2:27 PM

Note added by Georgiana.
Found the following log message:
- TCP: out of memory -- consider tuning tcp_mem

And there's an open issue about kubelet not monitoring the tcp mem of the node, but only disk ram and cpu and this is why node appeared as ready for more than 30m even though the tcp oom was happening
https://github.com/kubernetes/kubernetes/issues/62334.

Conclusion: The node ran out of tcp memory and kubelet wasn't equipped to find this early enough to kill the node and start out a new one. Issue is tracked upstream in kubernetes.

INCIDENT #262

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